

The bond enthalpies for six selected chemical bonds are shown below.

Bond	Bond enthalpy ΔH°_{298} (kJ/mol)
H—H	436
C—H	413
C=O	532
N—H	335
O—H	499
O=O	495

For a reaction in which 2 moles of $\text{H}_2(\text{g})$ reacts with 1 mole of $\text{O}_2(\text{g})$ to form 2 moles of $\text{H}_2\text{O}(\text{g})$, what is the heat of the reaction?

- ☐ A. 1367 kJ [13%]
☐ B. 369 kJ [28%]
☒ C. -629 kJ [54%]
☐ D. -1996 kJ [3%]

Correct

53% answered correctly

Explanation:

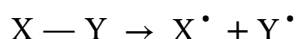
Calculating the heat of a reaction by Hess' law using bond enthalpies



According to **Hess' law**, if a reaction occurs in several steps, the **heat of the reaction** (overall enthalpy) is equal to the **sum of the enthalpy changes** for each step, because enthalpy is a **state function**. Accordingly, a chemical reaction can be viewed as a combined series of steps in which bonds within the reactants are broken as new bonds are formed to make the products.

Forming a bond is an **exothermic** process that releases energy (negative enthalpy), whereas **breaking a bond** is an **endothermic** process that requires energy (positive enthalpy). As a result, the enthalpies associated with breaking and forming a given type of bond are equal in magnitude but opposite in sign.

When a bond between two atoms is broken in a **homolytic cleavage**, the electrons within the bond are equally divided between the atoms.

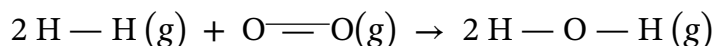
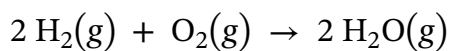


The **bond enthalpy** ΔH°_{298} (bond dissociation energy) is the energy needed to homolytically

break 1 mole of a given type of bond between two atoms in the **gas phase** at a temperature of 298 K. Applying Hess' law using measured bond enthalpies, the net heat of the reaction $\Delta H^\circ_{\text{rxn}}$ can be determined as the difference between the energy consumed to break bonds (ie, energy IN) and the energy released by making bonds (ie, energy OUT):

$$\Delta H^\circ_{\text{rxn}} = \text{Energy IN} - \text{Energy OUT}$$

In the reaction given in this question, 2 moles of (H–H) bonds are broken, 1 mole of O=O bonds is broken, and 4 moles of O–H bonds are formed.



Summing the bond enthalpies from the table for the broken bonds gives the total energy consumed:

$$\text{Energy IN} = \sum \left(\Delta H^\circ_{298} \text{ broken bonds} \right) = \left(2 \text{ mol (H} - \text{H)} \times 436 \frac{\text{kJ}}{\text{mol}} \right) + \left(1 \text{ mol (O} = \text{O)} \times 495 \frac{\text{kJ}}{\text{mol}} \right)$$

$$\text{Energy IN} = (872 \text{ kJ} + 495 \text{ kJ}) = 1367 \text{ kJ consumed}$$

Summing the bond enthalpies from the table for the formed bonds gives the total energy is released:

$$\text{Energy OUT} = \sum \left(\Delta H^\circ_{298} \text{ formed bonds} \right) = 4 \text{ mol (O} - \text{H)} \times 499 \frac{\text{kJ}}{\text{mol}} = 1996 \text{ kJ released}$$

Therefore, the net heat of the reaction $\Delta H^\circ_{\text{rxn}}$ is:

$$\Delta H^\circ_{\text{rxn}} = \text{Energy IN} - \text{Energy OUT}$$

$$\Delta H^\circ_{\text{rxn}} = 1367 \text{ kJ} - 1996 \text{ kJ} = -629 \text{ kJ}$$

(Choice A) 1367 kJ is only the energy required to break the bonds in the reactants.

(Choice B) 369 kJ is obtained if the number of moles of water formed is used in the calculation rather than the number of moles of O–H bonds formed.

(Choice D) –1996 kJ is only the energy released due to forming the O–H bonds in the products.

Educational objective:

Bond enthalpy is the energy needed to homolytically break 1 mole of a given type of bond between two atoms in the gas phase at 298 K. According to Hess' law, a reaction can be assessed as a series of endothermic bond-breaking and exothermic bond-forming steps, and the heat of the reaction is equal to the sum of the enthalpy changes for each step.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws